You are a senior Android developer and print-production automation engineer.

Build a native Android app for internal CALB department use. The app shall generate standardized CALB business cards in PDF format, including front side and back side, based on fixed corporate template rules. The first version only needs Android. No backend is required.

Project name: CALB Business Card Generator

Main Objective: Create an offline Android app that allows a user to input employee information, generate a CALB-standard business card, generate a static vCard QR code on the back side, preview both sides, and export print-ready PDF files that can be directly sent to a printing supplier.

Reference Design: Use the provided PDF template as the visual reference:

- File name: CALB 英文-名片模板-92.0x56.0-共2页.pdf
- Business card finished size: 92.0 mm × 56.0 mm
- Front side: CALB logo, slogan, company name, employee name, title, company line, mobile, email, postcode, address, hollow CALB watermark.
- Back side: centered static vCard QR code and caption “Scan and Save Contact”.
- Printing paper requirement: Ice White Pearl 300g.
- Corporate colors:
- Deep Blue: HEX #23496b, RGB 35, 73, 107
- Wise Grey: HEX #cedbea, RGB 206, 219, 234
- Keep the layout mostly fixed. Normal users should not freely drag or redesign the template.

Technology Requirements: 1. Native Android app. 2. Language: Kotlin. 3. UI: Jetpack Compose. 4. Architecture: MVVM. 5. Minimum SDK: 26 or above. 6. No cloud backend. 7. All data processing shall happen locally on the Android device. 8. Do not upload names, emails, phone numbers, or QR contents to any server. 9. Generate PDF locally. 10. Export PDF and share via Android share sheet.

Recommended Libraries:

- Jetpack Compose for UI.
- Android PdfDocument or another permissively licensed PDF generation approach.
- ZXing core or another permissively licensed QR generation library.
- Google libphonenumber for phone number parsing and E.164 formatting.
- Kotlin serialization or Moshi for local template configuration JSON.
- Android Storage Access Framework for saving files.
- Optional: Room or DataStore for local employee draft records.
- Use java.util.zip for ZIP export if batch export is implemented.

Avoid:

- No server dependency.
- No subscription or account system.
- No dynamic QR code in V1.
- No scan statistics in V1.
- No CRM integration in V1.
- No image/photo embedded into the vCard QR code.
- Do not use commercial PDF libraries unless clearly replaceable by an open/permissive alternative.

- Do not hard-code one employee’ s information as the only output.

Core Features V1:

1. Single Business Card Generation The app shall provide an input form with the following fields:

Required fields:

- English Name
- Last Name
- First Name
- Title
- Company Line
- Mobile Country
- Mobile Number
- Email

Optional/default fields:

- Website
- Postcode
- Street Address
- City
- State/Province
- Country
- Office Address Preset
- Chinese Name or Alias
- Department
- Note

Default company options:

- CALB Group Co., Ltd.
- CALB AMERICAS INC.
- CALB Americas Inc Allow admin/settings screen to edit company options, but normal generation should use dropdown selection.

Default address preset: Name: Brookshire Office Street: 839 FM 1489 Rd City: Brookshire State: TX Postcode: 77423 Country: United States Display address on card: 839 FM 1489 Rd, Brookshire, TX 77423, US

Default website: https://www.calb-tech.com

1. Phone Number Handling This is critical.

The app shall use libphonenumber or equivalent logic to normalize all phone numbers.

Example:

- User selects country: US
- User enters: 4015927928
- Display on business card: +1 401 592 7928 (US)

- vCard TEL value: +14015927928

Rules:

- The vCard phone number must be E.164 format.
- Do not include spaces, brackets, “US”, hyphens, or labels inside the vCard TEL value.
- The visual card may show a formatted display version with country label.
- Validate the number before allowing final export.
- If invalid, show clear error message.
- China example:
- Country: CN
- Input: 13800138000
- Display: +86 138 0013 8000 (CN)

- vCard: +8613800138000

- vCard QR Code Generate a static vCard QR code.

Use vCard 3.0 by default for compatibility.

vCard format must use CRLF line endings.

Example output: BEGIN:VCARD VERSION:3.0 N:Zhao;Alex;;; FN:Alex Zhao ORG:CALB AMERICAS INC. TITLE:Director of Pre-sale & Solution TEL;TYPE=CELL,WORK,VOICE:+14015927928 EMAIL;TYPE=INTERNET,WORK:alex.zhao@calb-tech.com ADR;TYPE=WORK:;;839 FM 1489 Rd;Brookshire;TX;77423;United States URL;TYPE=WORK:https://www.calb-tech.com NOTE:CALB Group Co., Ltd. END:VCARD

Important vCard rules:

- Include VERSION:3.0.
- Include both N and FN.
- Escape commas, semicolons, backslashes, and newlines according to vCard rules.
- Do not put display-only labels such as “(US)” into the TEL value.
- Do not include employee photos in the static QR.
- Do not make the QR too dense.
- QR error correction should default to M or Q.
- QR must preserve quiet zone.

- QR output for PDF should be high resolution or vector-like rendering.

- Business Card Front Side Layout Generate a PDF front page based on the fixed template.

Finished size:

- 92.0 mm × 56.0 mm

Print version with bleed:

- 3 mm bleed on all sides.
- PDF page size for print version: 98.0 mm × 62.0 mm.

- Keep the trim area centered inside the bleed page.

Preview version:

- 92.0 mm × 56.0 mm.
- No bleed.
- Used for internal review.

Coordinate system: Use millimeters in the template configuration, then convert to PDF points: 1 mm = 72 / 25.4 points.

Front side content:

- CALB logo in top-left area.
- CALB slogan under logo: Trust Efficient Win-Win.
- Top-right company text: CALB Group Co., Ltd.
- Name on left-middle.
- Title below name.
- Company line below title.
- Right-side information labels:
- Mobile
- Mail
- Postcode
- Address
- Right-side values aligned next to labels.
- Bottom hollow CALB watermark in Wise Grey or very light Deep Blue outline.

Suggested visual rules:

- Background: white.
- Main text color: Deep Blue #23496b.
- Watermark color: Wise Grey #cedbea with low opacity.
- Do not over-decorate.
- Keep layout consistent with the supplied PDF.

Field fitting rules:

- Name: large font. If too long, reduce font size.
- Title: allow up to 2 lines. If still too long, reduce font size.
- Email: reduce font size if long.
- Address: allow 2 lines.
- Do not allow text to overlap.

- If field cannot fit, show warning before export.

- Business Card Back Side Layout Generate PDF back page.

Back side content:

- White background.
- Centered static vCard QR code.
- Caption under QR: Scan and Save Contact

QR layout:

- QR size default: 26 mm to 30 mm.
- Center horizontally.
- Place vertically slightly above center, similar to the template.
- Caption centered below QR.
- Use black QR by default for best scan reliability.

- Optional setting: Deep Blue QR, but warn that black is safer for print.

- PDF Export The app shall support:

A. Preview PDF

- 2 pages.
- Page 1: front side.
- Page 2: back side.
- Size: 92.0 mm × 56.0 mm.
- No bleed.
- File name: CALB_Business_Card_{EnglishName}_Preview.pdf

B. Print-ready PDF

- 2 pages.
- Page 1: front side with 3 mm bleed.
- Page 2: back side with 3 mm bleed.
- Page size: 98.0 mm × 62.0 mm.
- Include trim/crop marks if practical.
- Embed fonts if possible.
- If font embedding is not possible, use bundled licensed fonts or Android system fonts consistently.
- File name: CALB_Business_Card_{EnglishName}_Print.pdf

C. Optional individual files:

- vCard .vcf file
- QR PNG or SVG file

- JSON record of the employee input

- Batch Generation Implement batch generation if feasible in V1. If full Excel XLSX is complex on Android, implement CSV import first.

CSV columns:
EnglishName,FirstName,LastName,Title,CompanyLine,MobileCountry,MobileNumber,Email,Website,Street,City,State,Postco

Batch workflow:

- User selects CSV file.
- App validates all rows.
- Show validation report.
- Rows with invalid phone/email/title length must be marked.

- User can export all valid cards.
- Output ZIP file containing:
- One preview PDF per person.
- One print PDF per person.
- One vCard file per person.
- One QR image per person.
- batch_validation_report.csv

ZIP folder structure: CALB_Business_Cards/ Alex_Zhao/ CALB_Business_Card_Alex_Zhao_Print.pdf CALB_Business_Card_Alex_Zhao_Preview.pdf Alex_Zhao.vcf Alex_Zhao_QR.png batch_validation_report.csv

1. Settings Screen Provide a simple settings screen for admin-like configuration.

Settings:

- Default company line.
- Default website.
- Address presets.
- QR size.
- QR color: black or Deep Blue.
- Enable/disable bleed.
- Default export mode.
- Template version.

- Allow editing template JSON only in advanced mode.

- Template Configuration Use a JSON template configuration instead of fully hard-coding all coordinates.

Example structure: { "page": { "finishedWidthMm": 92.0, "finishedHeightMm": 56.0, "bleedMm": 3.0 }, "colors": { "deepBlue": "#23496b", "wiseGrey": "#cedbea", "black": "#1e1e1e", "white": "#ffffff" }, "front": { "logo": {"x": 8.0, "y": 6.0, "w": 24.0, "h": 8.0}, "companyTop": {"x": 47.0, "y": 8.0, "fontSize": 7.0}, "name": {"x": 9.0, "y": 23.0, "fontSize": 12.0, "minFontSize": 8.5}, "title": {"x": 9.0, "y": 30.0, "fontSize": 5.8, "minFontSize": 4.8}, "companyLine": {"x": 9.0, "y": 36.0, "fontSize": 5.8}, "infoLabels": {"x": 47.0, "y": 30.0, "fontSize": 5.4}, "infoValues": {"x": 58.0, "y": 30.0, "fontSize": 5.4}, "watermark": {"x": 4.0, "y": 39.0, "w": 84.0, "h": 14.0, "opacity": 0.35} }, "back": { "qr": {"x": 33.0, "y": 12.0, "size": 27.0}, "caption": {"x": 0.0, "y": 42.5, "fontSize": 7.0, "align": "center"} } }

The actual coordinates may be adjusted after comparing the generated PDF with the provided reference PDF. Keep the JSON editable.

1. Assets The app should expect these assets:

2. CALB logo file, preferably SVG or high-resolution PNG.

3. Optional hollow CALB watermark asset, preferably SVG.
4. Optional font files, only if legally available for bundling.

If assets are missing:

- Show a clear warning.

- Use fallback text rendering for development.
- Do not crash.

- UI Screens Implement these screens:

A. Home Screen Buttons:

- Create Single Card
- Batch Import
- Settings
- About

B. Single Card Form Screen Input fields:

- English Name
- First Name
- Last Name
- Title
- Company Line dropdown
- Mobile Country dropdown
- Mobile Number
- Email
- Website
- Address preset dropdown
- Street
- City
- State
- Postcode
- Country
- Note

Actions:

- Validate
- Preview Front
- Preview Back
- Generate Preview PDF
- Generate Print PDF
- Share PDF

C. Preview Screen Show:

- Front side preview image.
- Back side preview image.
- QR code preview.
- vCard text preview in expandable section.

D. Batch Import Screen Show:

- Import CSV
- Validation result table

- Export valid rows
- Share ZIP

E. Settings Screen Show:

- Default company information
- Address presets
- QR options
- Export options
- Template version

F. About Screen Show:

- App name
- Version
- Internal-use disclaimer

- Privacy note: “All business card data is processed locally on this device. No personal information is uploaded.”

- Data Models Create Kotlin data classes:

EmployeeCardData:

- englishName: String
- firstName: String
- lastName: String
- title: String
- companyLine: String
- mobileCountryIso: String
- mobileRawInput: String
- mobileDisplay: String
- mobileE164: String
- email: String
- website: String
- street: String
- city: String
- state: String
- postcode: String
- country: String
- note: String?

AddressPreset:

- name
- street
- city
- state
- postcode
- country
- displayCountryShort

TemplateConfig:

- page
- colors
- front
- back

ValidationResult:

- isValid
- errors
- warnings
- Validation Rules Phone:
- Must be valid according to selected country.
- Must produce E.164 format.

Email:

- Must be syntactically valid.
- Warn if email domain is not calb-tech.com or configured allowed domain.

Title:

- Required.
- Warn if too long for the template.

Name:

- Required.
- Warn if too long.

Address:

- Required for print card, unless user selects "hide address".
- Warn if longer than 2 lines.

Company:

- Required.
- PDF Rendering Requirements Implement a PDF rendering service with functions:

generatePreviewPdf(data: EmployeeCardData): Uri/File generatePrintPdf(data: EmployeeCardData): Uri/File generateBatchZip(dataList: List<EmployeeCardData>): Uri/File renderFrontPage(canvas, data, config, isPrintMode) renderBackPage(canvas, data, config, isPrintMode)

PDF rendering details:

- Use mm-to-point conversion.
- For print mode, add bleed offset of 3 mm.
- Keep trim area centered.
- Draw all text using Deep Blue.
- Draw watermark in light Wise Grey.
- Draw QR with sufficient resolution.
- Keep white quiet zone around QR.
- Avoid rasterizing the whole page if possible.
- If rasterization is used for preview, final PDF text should still be sharp if feasible.
- QR Rendering Requirements Implement a QR service with functions:

buildVCard(data: EmployeeCardData): String generateQrBitmap(vCard: String, sizePx: Int): Bitmap generateQrForPdf(vCard: String, targetSizeMm: Float): Drawable/Bitmap

QR requirements:

- Use UTF-8.
- Use error correction M by default.
- Use quiet zone.
- Test QR code by saving PNG.
- Ensure QR can be scanned from printed PDF.
- Example Test Data Use this as default demo data:

English Name: Alex Zhao First Name: Alex Last Name: Zhao Title: Director of Pre-sale & Solution Company Line: CALB AMERICAS INC. Mobile Country: US Mobile Number: 4015927928 Email: alex.zhao@calb-tech.com Website: https://www.calb-tech.com Street: 839 FM 1489 Rd City: Brookshire State: TX Postcode: 77423 Country: United States Note: CALB Group Co., Ltd.

Expected display phone: +1 401 592 7928 (US)

Expected vCard phone: +14015927928

Expected vCard: BEGIN:VCARD VERSION:3.0 N:Zhao;Alex;;; FN:Alex Zhao ORG:CALB AMERICAS INC. TITLE:Director of Pre-sale & Solution TEL;TYPE=CELL,WORK,VOICE:+14015927928 EMAIL;TYPE=INTERNET,WORK:alex.zhao@calb-tech.com ADR;TYPE=WORK:;;839 FM 1489 Rd;Brookshire;TX;77423;United States URL;TYPE=WORK:https://www.calb-tech.com NOTE:CALB Group Co., Ltd. END:VCARD

1. Testing Requirements Add unit tests for:
2. US phone number normalization.
3. China phone number normalization.
4. Invalid phone number detection.

5. Email validation.
6. vCard escaping.
7. vCard generation.
8. mm-to-point conversion.
9. PDF file creation.
10. QR generation.
11. Batch CSV validation.

Add manual QA checklist:

- Scan generated QR with iPhone camera.
- Scan generated QR with Samsung camera.
- Scan generated QR with WeChat.
- Confirm contact save page shows:
- Name
- Company
- Title
- Mobile number
- Email
- Address
- Website
- Confirm US number shows as +1 number.
- Print sample at actual size and scan it.
- Confirm PDF physical size is 92 mm × 56 mm for preview.
- Confirm print PDF has 3 mm bleed.
- Deliverables Generate a complete Android Studio project with:
- build.gradle files.
- Kotlin source code.
- Jetpack Compose UI.
- PDF generation service.
- QR generation service.
- vCard generation service.
- Phone validation service.
- CSV batch import/export.
- Sample template JSON.
- Sample asset placeholders.
- README.md.
- Manual test checklist.
- Example CSV file.
- README Requirements The README shall explain:
- How to build the app.
- How to run it on Android Studio.
- How to import CSV.

- How to generate single card PDF.
- How to generate print-ready PDF.
- How phone number normalization works.
- How to replace CALB logo assets.
- How to adjust the template JSON.

- Known limitations.

- Acceptance Criteria The project is acceptable only if:

- The app builds successfully in Android Studio.

- User can enter employee information and generate a two-page PDF.
- Front side resembles the reference CALB card layout.
- Back side contains a scannable static vCard QR code.
- US phone number is correctly stored in vCard as +14015927928 when input is 4015927928 and country is US.
- Exported preview PDF is 92 mm × 56 mm.
- Exported print PDF is 98 mm × 62 mm with 3 mm bleed.
- PDF can be shared from Android.
- All personal data remains local.
- Batch CSV import can validate and export multiple cards, or if not fully implemented, the app must include a clear TODO and working single-card export.

Please implement the app step by step. Start with the project structure, data models, vCard generation, phone normalization, QR generation, then PDF generation, then UI screens, then batch import/export.